



# Backplane & Motherboard Design

PCBs that carry other PCBs — the highest-density wiring

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Large-format boards for industrial systems, servers, AI hardware

Audience: Senior PCB engineers + hardware architects

Topology • Connectors • SI • Power • Layer counts



# What is a Backplane / Motherboard

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- ▶ Backplane: PCB whose primary purpose is connecting OTHER PCBs (daughter cards)
- ▶ Motherboard: similar, but typically also hosts main CPU + chipset (PC use)
- ▶ Carries: power, ground, signals between cards via connectors
- ▶ Common topologies: bus (parallel), star (point-to-point), mesh
- ▶ Examples: PC motherboard, server blades, industrial PLC racks, FPGA dev boards
- ▶ Size: large (200-500 mm) — special fab handling required
- ▶ Layer count: 12-30+ layers for signal integrity + power delivery

# Typical Topologies

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### Bus topology (legacy)

- ▶ Daughter cards connect to common bus
- ▶ Parallel ATA, ISA, VME used this
- ▶ Simple, low cost
- ▶ Signal integrity issues at high speed
- ▶ Limited bandwidth (shared bus)
- ▶ Mostly obsolete except for very-low-cost

### Switched / point-to-point

- ▶ Backplane is a SWITCH or hub
- ▶ Each card-to-card connection is independent
- ▶ PCIe, Ethernet, custom serdes
- ▶ Much higher bandwidth than bus
- ▶ More complex, more layers
- ▶ Modern standard

# Connector Selection

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- ▶ Backplane connectors are HIGH-RELIABILITY + HIGH-INSERTION
- ▶ Common: Hard Metric (DIN 41612), VITA 41/42/46, MicroTCA, PCIe edge
- ▶ Pin counts: 100-700 per connector typical; some > 1000
- ▶ Differential pairs supported (USB, PCIe, Ethernet) at GHz speeds
- ▶ Insertion force: high; insertion cycles: 100-1000+
- ▶ Press-fit pins (no soldering required for backplane side)
- ▶ Modern: high-speed connectors from Molex, TE Connectivity, Amphenol
- ▶ Cost: ₹500-5000 per connector at low volume

# Layer Counts + Stackup

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- ▶ Backplane typical: 12-20 layers
- ▶ Server motherboard: 20-30 layers
- ▶ AI accelerator + GPU: 30-40 layers
- ▶ Plane allocation: dedicated GND and power planes (often 30%+ of layer budget)
- ▶ Signal layers: organized in pairs (GND-signal-GND sandwich for SI)
- ▶ Stripline routing for sensitive signals (PCIe, DDR)
- ▶ Cost: 30-100× standard 4-layer due to layer count alone
- ▶ Lead time: 4-8 weeks specialty

# Signal Integrity Considerations

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- ▶ Long trace lengths (>250 mm typical) → loss budget critical
- ▶ Use low-loss material (Megtron 6 / Tachyon) for traces > 100 mm at GHz
- ▶ Back-drilling: remove via stubs (essential for PCIe Gen 4+)
- ▶ Connector launch design: trace transitions critical for SI
- ▶ AC coupling at every TX (USB, PCIe) at the right end of the trace
- ▶ Equalization in transmitters + receivers (modern SERDES)
- ▶ Pre-layout SI simulation (HyperLynx, ADS) MANDATORY
- ▶ Post-layout sim with extracted parasitics

# Power Delivery Network (PDN)

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- ▶ Backplanes carry hundreds of amps total across all rails
- ▶ Multiple voltage rails: 12V, 5V, 3.3V, 1.2V, 0.9V, multiple analog rails
- ▶ Heavy Cu (2-4 oz) on power planes
- ▶ PoL (Point-of-Load) regulators per card (vs centralised) to reduce IR drop
- ▶ Bulk capacitance at every card slot + backplane regulator
- ▶ PDN impedance target  $< 5 \text{ m}\Omega$  at GHz (for VDDQ on DDR rails)
- ▶ Multiple parallel power planes for split rails (e.g., VDDQ\_A vs VDDQ\_B)
- ▶ Power-sequencing IC for orderly startup of multi-rail system



# Manufacturing Challenges

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- ▶ Board size 500+ mm — only specialty fabs can handle
- ▶ Layer count 20+ — complex pressing, controlled-impedance test on every panel
- ▶ Back-drilling: time-consuming + expensive (\$)
- ▶ Press-fit insertion: precision tooling for connector pins
- ▶ Inspection: AOI + X-ray + boundary-scan + functional test
- ▶ Yield: lower than standard PCBs (more steps = more error opportunities)
- ▶ Repair: micro-rework on > 20-layer PCB is extremely hard
- ▶ Cost: ₹50k-500k+ per board at low volume

# Common Backplane Standards

## VPX / VITA 46

Military / aerospace  
High-speed serial

## MicroTCA

Telecom / instrumentation  
PCIe-based

## CompactPCI

Industrial legacy  
Updates with serial CPCI

## PCIe (consumer)

Desktop motherboards  
Gen 3/4/5 serdes

## OCP (Open Compute)

Datacenter servers  
Air + liquid cooling

## AMC (Advanced Mezzanine)

Modular system board  
for telecom

# Mechanical Design

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- ▶ Backplane fits in 19" rack (most industrial / server systems)
- ▶ Front-loading vs rear-loading card insertion
- ▶ Card guides for accurate alignment to backplane connectors
- ▶ Hot-plug support: card insertion / removal while power is on (PCIe hot-plug, hot-swap)
- ▶ Power management: ramp regulators in software to prevent inrush damage
- ▶ Cooling: significant airflow needed across the backplane + cards
- ▶ Common bracket: lock mechanism + thumbscrew for service access
- ▶ ESD bonding: all chassis brackets bonded together

# Verification + Test

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- ▶ Pre-fab SI simulation (HyperLynx, ADS) at every speed grade
- ▶ Post-fab TDR (Time Domain Reflectometry) on every controlled-impedance trace
- ▶ Eye diagram capture at every high-speed link end
- ▶ Power-rail measurement under load (multimeter + scope ripple)
- ▶ Burn-in run with all daughter cards installed
- ▶ EMC compliance scan with all cards in
- ▶ Thermal imaging during burn-in
- ▶ Connector insertion-cycle test (1000+ cycles typical for press-fit)

# Cost + Vendor Options

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- ▶ Backplane fab vendors: TTM Technologies, Sanmina, Unimicron, AT&S
- ▶ Cost: ₹50k-500k+ per board at low volume
- ▶ At volume (1000+): ₹10-50k per board
- ▶ Lead time: 4-8 weeks for prototype, 8-12 for production
- ▶ Indian options: Centum, AT&S India for specialty multilayer
- ▶ Talk to vendor early — design rules vary
- ▶ Get multiple quotes — variance can be 2-3×

# Key Takeaways

Key takeaways from this lecture

## Specialty design

20-40 layer is normal here.

## Modern = point-to-point

No more parallel buses.

## Connectors are critical

Press-fit, high reliability.

## PDN at GHz

< 5 mΩ impedance target.

## SI sim mandatory

HyperLynx / ADS at all speed grades.

## Cost + lead time

Plan months ahead.

# Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

## References & Further Learning

- ▶ VITA / PICMG standards — backplane specs
- ▶ Open Compute Project (OCP) — server backplane
- ▶ Eric Bogatin / Stephen Hall — SI for backplanes
- ▶ HyperLynx / Sigrity / ADS — SI tools
- ▶ TTM Technologies / Sanmina — vendor capability
- ▶ DesignCon papers on backplane design

Thank you • Build something this week.